Product Specification

■ **Preliminary Specification**
□ **Final Specification**

Module	21.5 Inch Color TFT-LCD with Touch
Model Name	G215HAT02.7
TP FW version	TBD

Company		**Approved by**	**Date**
		Elsie Kuo	2026/1/5
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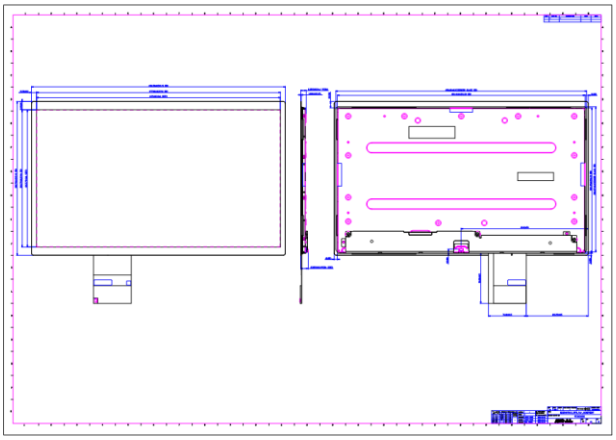

</td></tr>
</table>

1. Operating Precautions

1) Since front polarizer is easily damaged, please be cautious and not to scratch it.
2) Be sure to turn off power supply when inserting or disconnecting from input connector.
3) Wipe off water drop immediately. Long contact with water may cause discoloration or spots.
4) When the panel surface is soiled, wipe it with absorbent cotton or soft cloth.
5) Since the panel is made of glass, it may be broken or cracked if dropped or bumped on hard surface.
6) To avoid ESD (Electro Static Discharde) damage, be sure to ground yourself before handling TFT-LCD Module.
7) Do not open nor modify the module assembly.
8) Do not press the reflector sheet at the back of the module to any direction.
9) In case if a module has to be put back into the packing container slot after it was taken out from the container, do not press the center of the LED light bar edge. Instead, press at the far ends of the LED light bar edge softly. Otherwise the TFT Module may be damaged.
10) At the insertion or removal of the Signal Interface Connector, be sure not to rotate nor tilt the Interface Connector of the TFT Module.
11) TFT-LCD Module is not allowed to be twisted & bent even force is added on module in a very short time. Please design your display product well to avoid external force applying to module by end-user directly.
12) Small amount of materials having no flammability grade is used in the LCD module. The LCD module should be supplied by power complied with requirements of Limited Power Source (IEC60950 or UL1950), or be applied exemption.
13) Severe temperature condition may result in different luminance, response time and lamp ignition voltage.
14) Continuous operating TFT-LCD display under low temperature environment may accelerate lamp exhaustion and reduce luminance dramatically.
15) The data on this specification sheet is applicable when LCD module is placed in landscape position.
16) Continuous displaying fixed pattern may induce image sticking. It’s recommended to use screen saver or shuffle content periodically if fixed pattern is displayed on the screen.

2. General Description

This specification applies to the 21.5 inch-wide Color TFT-LCD Module with Touch G215HAT02.7. The display supports the Full HD-1920(H)x1080(V) screen format and 16.7M colors. All input signals are LVDS interface comaptible.

2.1 Display Characteristics (G215HAT02.7)

The following items are characteristics summary on the table under 25 ℃ condition:

Items	**Unit**	**Specifications**
Screen Diagonal	[mm]	(21.5")
Active Area	[mm]	476.064 (H) x 267.786 (V)
Pixels H x V		1920(x3) x 1080
Pixel Pitch	[um]	247.95 (per one triad) ×247.95
Pixel Arrangement		R.G.B. Vertical Stripe
Display Mode		AHVA Mode, Normally Black
White Luminance (Center)	[cd/m^2]	350 cd/m^2 (Typ.)
Contrast Ratio		1000 (Typ.)
Optical Response Time	[msec]	22 (Tr + Tf)
Nominal Input Voltage	[Volt]	5 V(Typ)
Power Consumption (VDD line + LED line)	[Watt]	LCD module: PDD (Typ.)= 2.5 @ White pattern, Fv=60Hz Backlight unit: PBLU=(10.03)W (Typ.)
Weight	[Grams]	TBD g(Typ.)
Physical Size (TTL)	[mm]	496.5 x 300.79 x 15.85 (excluded LED driver board)
Electrical Interface		eDP 1.2
Support Color		16.7M
Surface Treatment		AS coating
Temperature Range Operating Storage (Non-Operating)	 [°C] [°C]	 0 to +60 -20 to +60
RoHS Compliance		RoHS Compliance

2.2 General Touch Characteristics

Item		Unit	Specifications
Type			Projected Capacitive Touch Panel
Structure			Glass / Glass
Mounting type			Direct Bonding
Input Mode			Multi Finger
Cover lens	O.D.	[mm]	496.5 x 300.79
	Thickness	[mm]	1.1
C/L Visual Area		[mm]	477.06 x 268.79
Sensor Glass	O.D.	[mm]	490.62 x 284.27
	Thickness	[mm]	0.7
TP Active Area		[mm]	479.66 x 271.39
Substrate Material			SDL CS Glass
Touch Controller			ILI 2521
Touch points			10
Transmittance (%)		%	≥ 87
Single Touch Point Life Time			1 Millions
Surface Hardness			7H
Interface			USB 2.0 full speed
Single / Multi-touch Accuracy		[mm]	Center:+/-1.5mm Edge: +/-2 mm
Linearity		[mm]	Center: +/-1.5mm Edge: +/-2mm
The smallest distance between 2 points (each point diameter is 7mm)		[mm]	TBD
Channel (X * Y)			TBD
Report Rate (points /sec)		[Hz]	>100
Power Consumption		[mW]	TBD
Response Time		[ms]	TBD
Operating System			Support Win8 / Win10 ,Linux & Android
TP Firmware Version			TBD

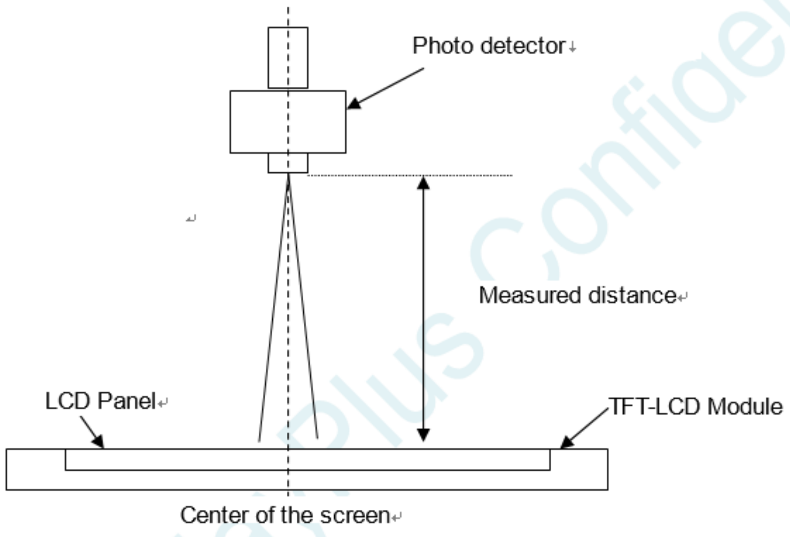
2.3 Optical Characteristics

The optical characteristics are measured under stable conditions at 25℃ (Room Temperature):

Item	Unit	Conditions	Min.	Typ.	Max.	Note
Viewing Angle	[degree]	Horizontal (Right) CR = 10 (Left)		89 89	- -	1, 2
	[degree]	Vertical (Upper) CR = 10 (Lower)		89 89	- -	
Contrast ratio		Normal Direction	600	1000	-	3
Response Time	[msec]	Tr + Tf		22		4
Color / Chromaticity Coordinates (CIE)		Red x	0.626	0.656	0.686	5
		Red y	0.298	0.328	0.358	
		Green x	0.263	0.293	0.323	
		Green y	0.584	0.614	0.644	
		Blue x	0.117	0.147	0.177	
		Blue y	0.037	0.067	0.097	
Color Coordinates (CIE) White		White x	0.283	0.313	0.343	
		White y	0.299	0.329	0.359	
Central Luminance	[cd/m^2]		280	350		6
Luminance Uniformity	[%]		80	85	-	7
Color Gamut	[%]	sRGB	95%	99%		

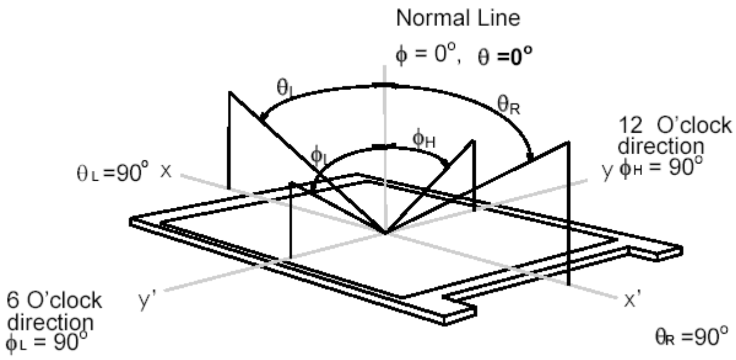
Note 1: Measurement method

Before measuring, the LCD module should be turn on 30 minutes at room temperature. In order to stabilize the luminance, the measurement should be executed after lighting Backlight for 30 minutes in a stable, windless and dark room.



Note 2: Definition of viewing angle measured by ELDIM (EZContrast 88)

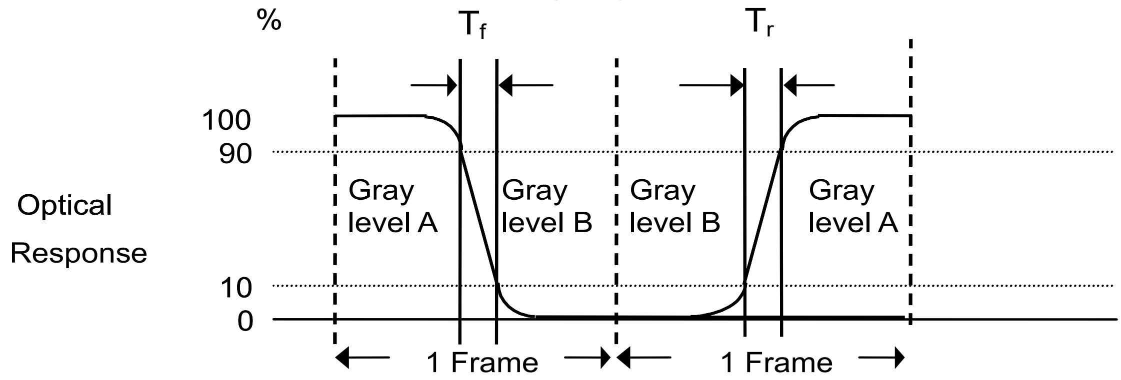
Viewing angle is the measurement of contrast ratio ≧ 10, at the screen center, over a 180° horizontal and 180° vertical range (off-normal viewing angles). The 180° viewing angle range is broken down as follows; 90° (θ) horizontal left and right and 90° (Φ) vertical, high (up) and low (down). The measurement direction is typically perpendicular to the display surface with the screen rotated about its center to develop the desired measurement viewing angle.



Note 3: Contrast ratio is measured by TOPCON SR-3

Note 4: Response time measurement

The output signals of photo detector are measured when the input signals are changed from "Gray level A" to "Gray level B" (falling time, TF), and from "Gray level B" to "Gray level A" (rising time, TR), respectively. The response time is interval between the 10% and 90% of optical response.



The gray to gray response time is defined as the following table.

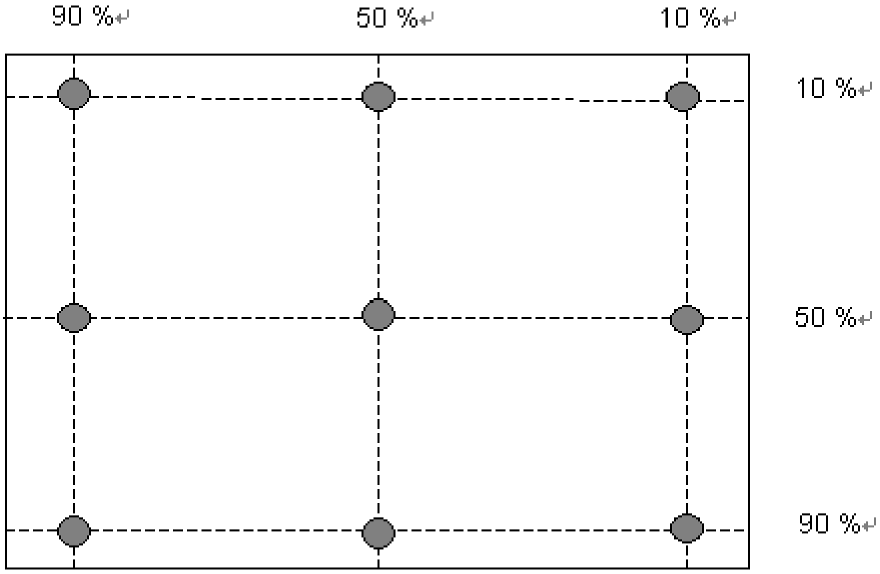
Gray Level to Gray Level		***Target gray level***				
		L0	***L63***	***L127***	***L191***	***L255***
Start gray level	***L0***					
	L63					
	L127					
	L191					
	L255					

- T_{GTG_typ} is the total average time at rising time and falling time of gray to gray.

Note 5: Color chromaticity and coordinates (CIE) is measured by TOPCON SR-3

Note 6: Central luminance is measured by TOPCON SR-3

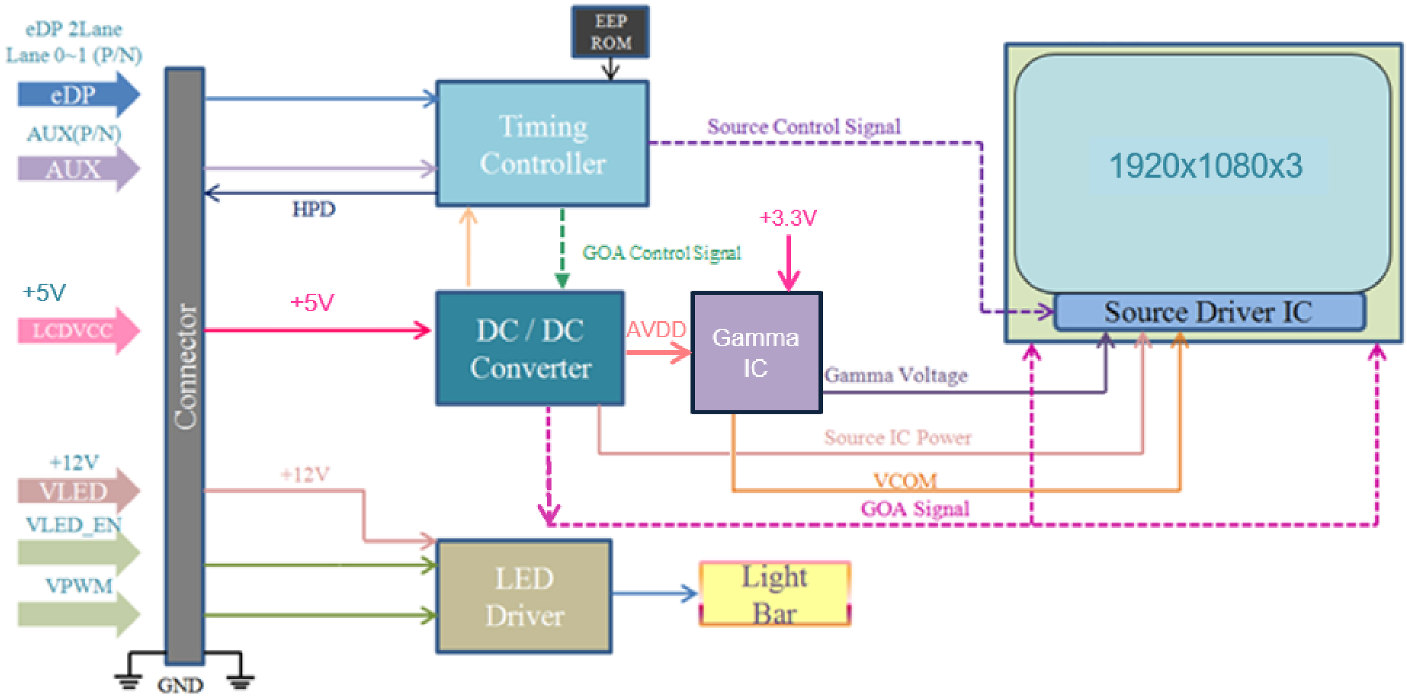
Note 7: Luminance uniformity of these 9 points is defined as below and measured by TOPCON SR-3



$$\text{Uniformity} = \frac{\text{Minimum Luminance in 9 points}(1-9)}{\text{Maximum Luminance in 9 Points}(1-9)}$$

3. Functional Block Diagram

The following diagram shows the functional block of this model.



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4. Absolute Maximum Ratings

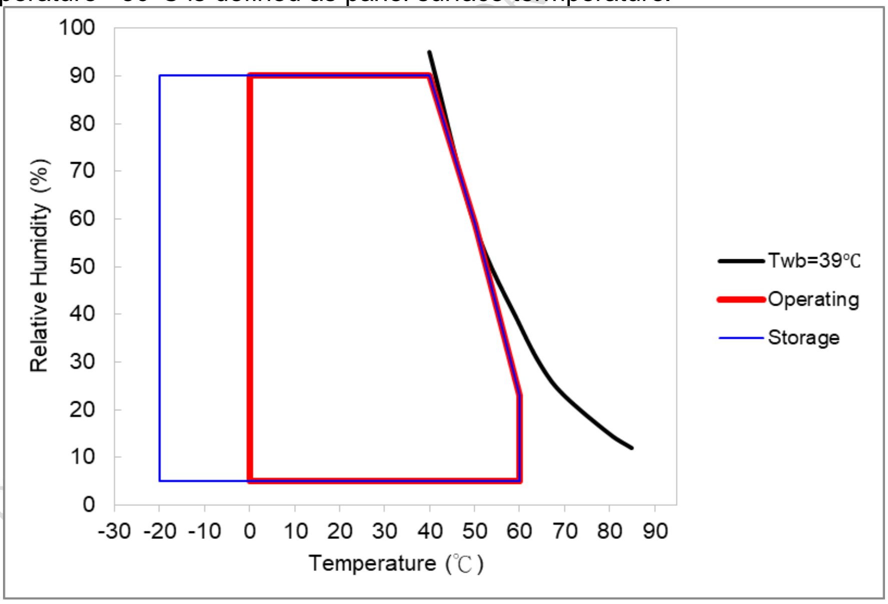
4.1 TFT LCD Module

Item	Symbol	Min	Max	Unit	Conditions
Logic/LCD Drive Voltage	VDD	-0.3	+5.5	[Volt]	Note 1,2

4.2 Absolute Ratings of Environment

Item	Symbol	Min.	Max.	Unit	Conditions
Operating Temperature	TOP	0	60	[ºC]	Note 3 & 4
Operation Humidity	HOP	5	90	[%RH]	
Storage Temperature	TST	-20	60	[ºC]	
Storage Humidity	HST	5	90	[%RH]	

Note 1: With in Ta (25 ºC)
Note 2: Permanent damage to the device may occur if exceeding maximum values
Note 3: For quality perfermance, please refer to AUO IIS(Incoming Inspection Standard).
Note 4: Operation Temperature +50°C is defined as panel surface termperature.



5. Electrical Characteristics

5.1 TFT LCD Module

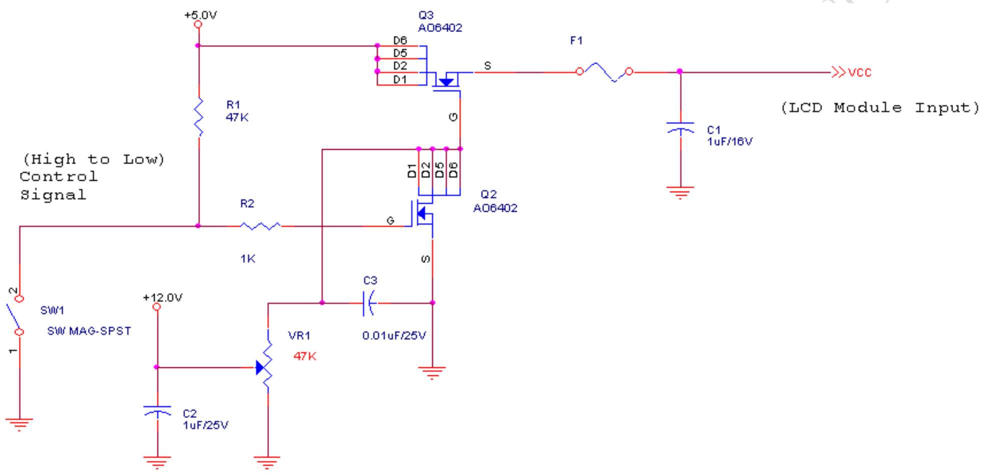
5.1.1 Power Specification

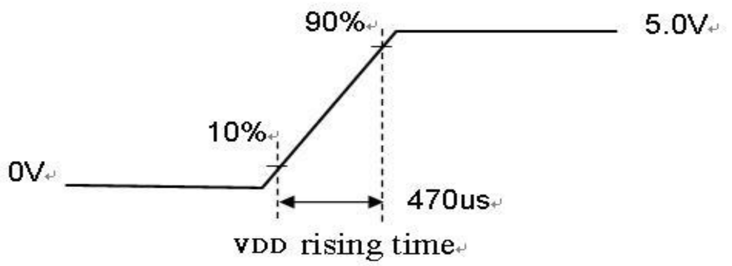
Input power specifications are as follows:

Symbol	Description	Min	Typ	Max	Unit	Remark
VDD	Input voltage	4.5	5	5.5	[Volt]	
IDD	Input Current (RMS)	-	0.4	0.43	[A]	VDD= 5.0V, White Pattern, Fv=60Hz
PDD	Consumption	-	2.0	2.15	[Watt]	VDD= 5.0V, White Pattern, Fv=60Hz
IRush	Inrush Current	-	-	2	[A]	Note 5-1
VDDrp	Allowable VDD Ripple Voltage	-	-	100	[mV]	VDD= 5.0V, White Pattern, Fv=60Hz

Note 1: Measurement conditions:

The duration of rising time of power input is 470us.



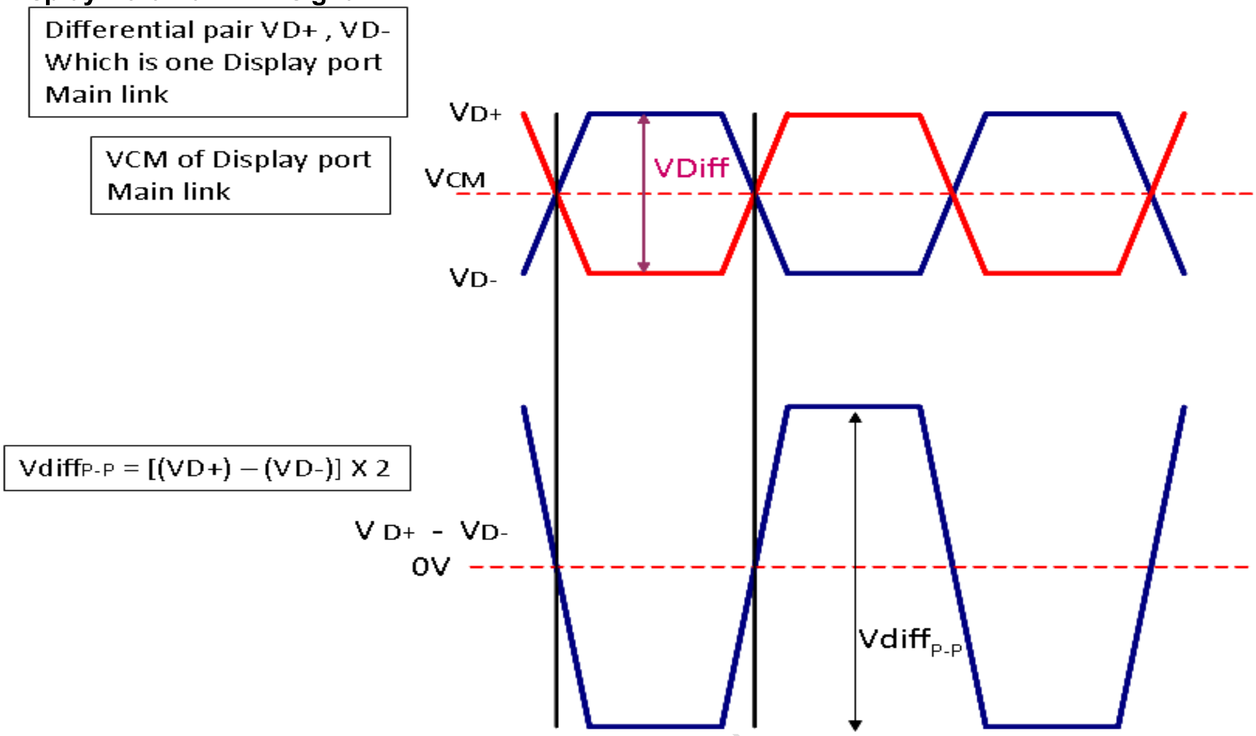


5.1.2 Signal Electrical Characteristics

Input signals shall be low or High-impedance state when VDD is off.

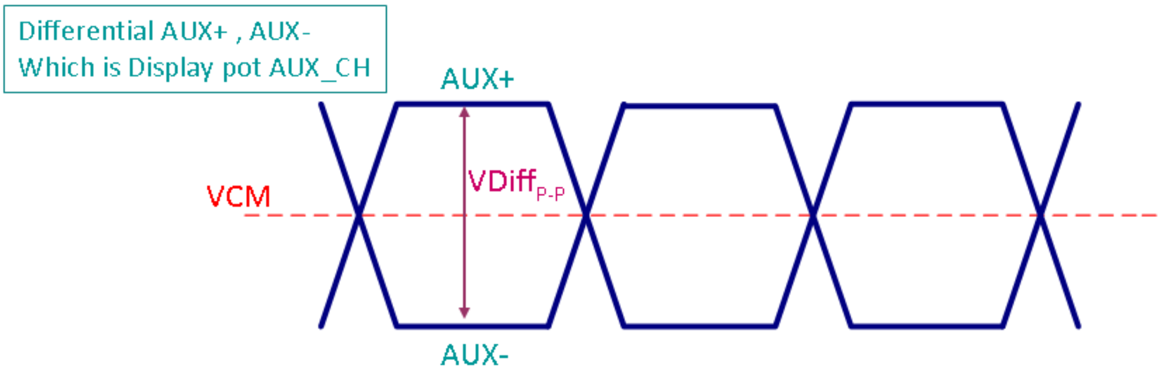
Signal electrical characteristics are as follows;

Display Port main link signal:



Display port main link					
		Min	Typ	Max	unit
VCM	RX input DC Common Mode Voltage		0		V
$VDiff_{P-P}$	Peak-to-peak Voltage at a receiving Device	150		1320	mV

Display Port AUX_CH signal:



Display port AUX_CH					
		Min	Typ	Max	unit
VCM	AUX DC Common Mode Voltage		0		V
$VDiff_{P-P}$	AUX Peak-to-peak Voltage at a receiving Device	270		800	mV

Follow as VESA display port standard

Display Port VHPD signal:

Display port VHPD					
		Min	Typ	Max	unit
VHPD	HPD Voltage (input impedance 1KΩ)	2.25	-	2.75	V

5.2 Backlight Unit

5.2.1 LED Driver

Following characteristics are measured under a stable condition at 25 ºC (Room Temperature):

I_F	LED Forward Current		55		mA	Ta = 25 ºC
V_F	LED Forward Voltage	-	2.85	3.3	Volt	I_F =55mA, Ta = 25ºC
P_{LED}	LED Power Consumption	-	10.03	11.62	Watt	I_F =55mA, Ta = 25ºC
LED Life Time		50,000			Hrs	I_F =55mA, Ta= 25ºC

Note 1: Ta means ambient temperature of TFT-LCD module.

Note 3: I_F, V_F , P_{LED} are defined for single LED.

Note 4: If G215HAY02.7 module is driven by high current or at high ambient temperature & humidity condition. The LED life will be reduced.

Note 5: Definition of life time: LED brightness becomes 50% of its original value. The minimum life time of LED unit is defined at the condition of I_F =55mA, Ta = 25ºC (Room temperature)

Note 6: Each LED light bar consists of 64 pcs LED package (4 strings x 16 pcs / string)

5.3 Touch Sensor Module

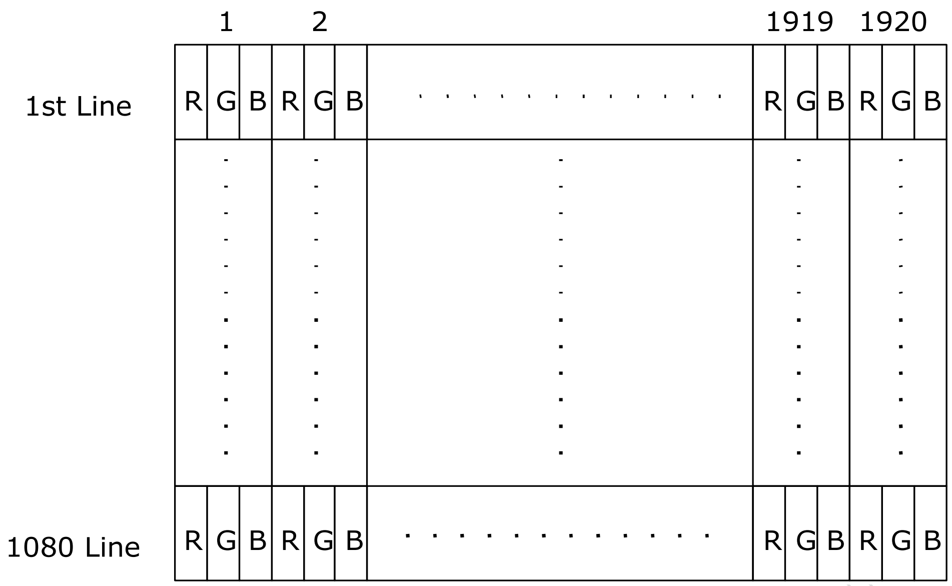
5.3.1 Power Specification

Items	Symbol	Specifications			Unit	Notes
		Min.	Typ.	Max.		
Touch sensor module Power Supply	TP_VDD	4.5	5	5.5	V	
Touch sensor module Power Consumption	PTP Active	-	475	660	mW	
Touch Sensor Module Power ripple	TP_VDDrp	-	-	100	mV	

6. Signal Characteristic

6.1 Pixel Format Image

Following figure shows the relationship of the input signals and LCD pixel format.



6.2 Signal Description

6.2.1 TFT LCD Module Connector Description

Connector Name / Designation	For Signal Connector
Manufacturer	IPEX or compatible
Type / Part Number	IPEX 20765-030E-11A
Mating Housing/Part Number	IPEX 20704-030-## or compatible

6.2.2 TFT LCD Module Pin Assignment

PIN #	SIGNAL NAME	DESCRIPTION
1	NC	No connect
2	VLED_IN	Backlight power
3	VLED_IN	Backlight power
4	VLED_IN	Backlight power
5	VLED_IN	Backlight power
6	NC	No connect
7	NC	No connect
8	VLED_PWM	System PWM signal Input
9	VLED_EN	Backlight On / Off
10	BL_GND	Backlight ground
11	BL_GND	Backlight ground
12	BL_GND	Backlight ground
13	BL_GND	Backlight ground
14	HPD	HPD signal pin
15	LCD_GND	LCD logic and driver ground
16	LCD_GND	LCD logic and driver ground
17	LCD_Self_Test	LCD Panel Self Test Enable
18	LCD_VCC	LCD logic and driver power
19	LCD_VCC	LCD logic and driver power
20	H_GND	High Speed Ground
21	AUX_CH_N	Comp Signal Auxiliary Ch.
22	AUX_CH_P	True Signal Auxiliary Ch.
23	H_GND	High Speed Ground
24	Lane0_P	True Signal Link Lane 0
25	Lane0_N	Comp Signal Link Lane 0
26	H_GND	High Speed Ground
27	Lane1_P	True Signal Link Lane 1
28	Lane1_N	Comp Signal Link Lane 1
29	H_GND	High Speed Ground
30	PANEL_VDD_DET	Ground, Panel LCD_VDD=5V

6.2.3 TP Connector Description

Connector Name / Designation	For Signal Connector
Manufacturer	ACES
Type / Part Number	50224-00501-001

6.2.4 TP PIN assignment(USB Interface)

USB Interface	
Pin No.	Designation
1	GND
2	VDD
3	GND
4	D+
5	D-

6.3 Timing Characteristics

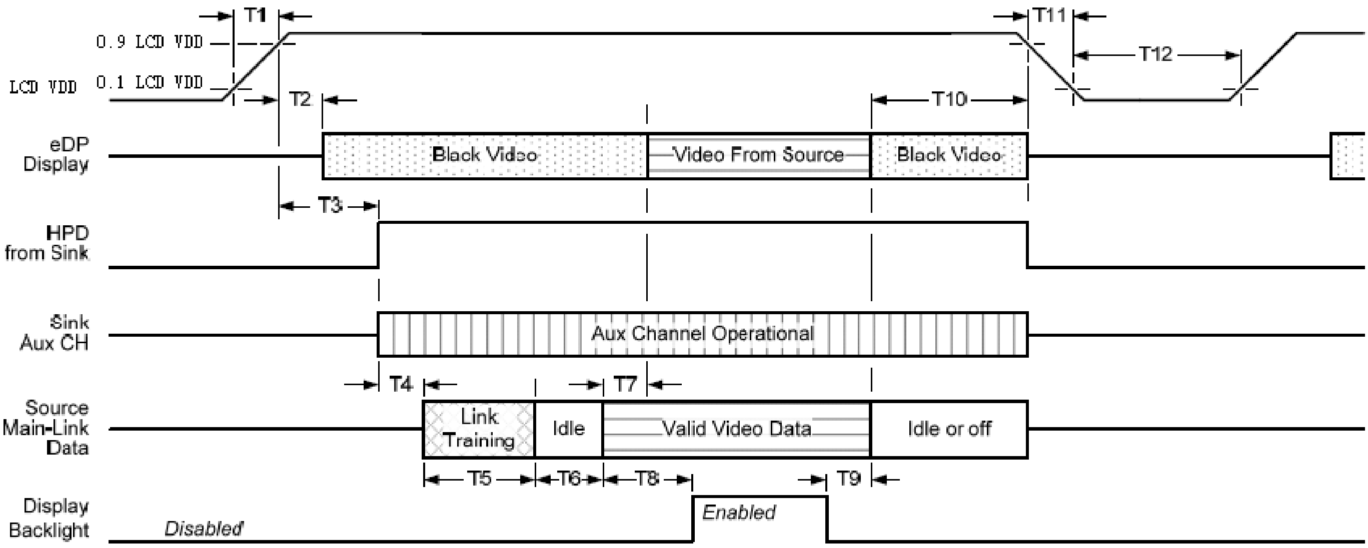
Basically, interface timings should match the 1920x1080 /60Hz manufacturing guide line timing.

Parameter		Symbol	Min.	Typ.	Max.	Unit
Frame Rate		-		60		Hz
Clock frequency		$1/T_{Clock}$		143.2		MHz
Vertical Section	Period	T_V		1130		T_{Line}
	Active	T_{VD}		1080		
	Blanking	T_{VB}		50		
Horizontal Section	Period	T_H		2100		T_{Clock}
	Active	T_{HD}		1920		
	Blanking	T_{HB}		180		

6.4 Power ON/OFF Sequence

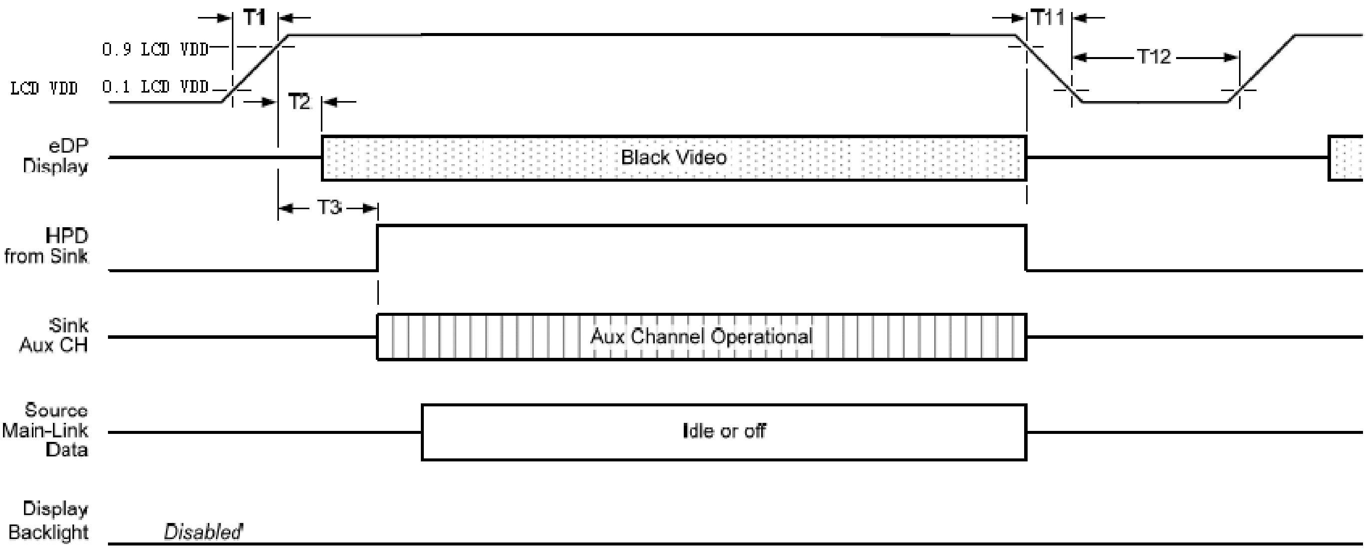
Power on/off sequence is as follows. Interface signals and LED on/off sequence are also shown in the chart. Signals from any system shall be Hi-Z state or low level when VDD is off

Display Port panel power sequence:



Display port interface power up/down sequence, normal system operation

Display Port AUX_CH transaction only:



Display port interface power up/down sequence, AUX_CH transaction only

Display Port panel power sequence timing parameter:

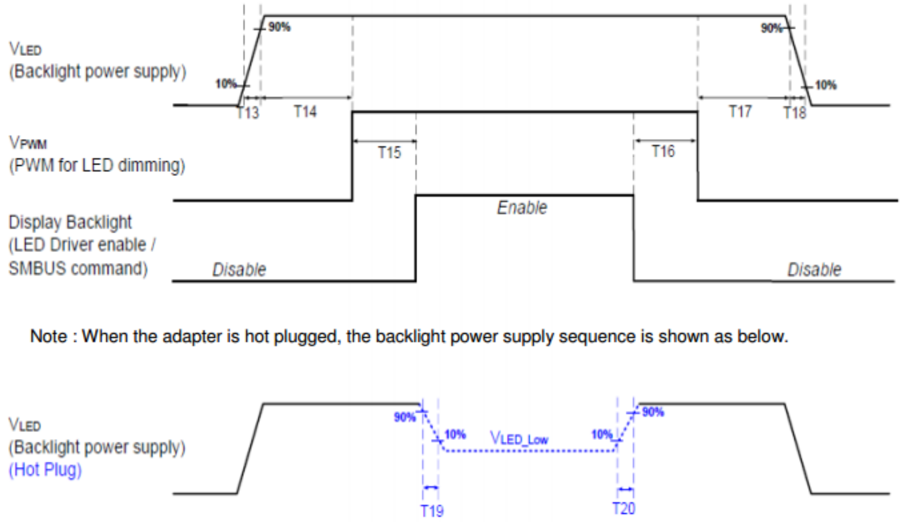
Timing parameter	Description	Reqd. by	Limits			Notes
			Min.	Typ.	Max.	
T1	power rail rise time, 10% to 90%	source	0.5ms		10ms	
T2	delay from LCDVDD to black video generation	sink	0ms		200ms	prevents display noise until valid video data is received from the source
T3	delay from LCDVDD to HPD high	sink	0ms		200ms	sink AUX_CH must be operational upon HPD high.
T4	delay from HPD high to link training initialization	source				allows for source to read link capability and initialize.
T5	link training duration	source				dependant on source link to read training protocol.
T6	link idle	source				Min accounts for required BS-Idle pattern. Max allows for source frame synchronization.
T7	delay from valid video data from source to video on display	sink	0ms		50ms	max allows sink validate video data and timing.
T8	delay from valid video data from source to backlight enable	source				source must assure display video is stable.
T9	delay from backlight disable to end of valid video data	source				source must assure backlight is no longer illuminated.
T10	delay from end of valid video data from source to power off	source	0ms		500ms	
T11	power rail fall time, 905 to 10%	source			10ms	
T12	power off time	source	500ms			

Note1: The sink must include the ability to generate black video autonomously. The sink must automatically enable black video under the following conditions:
-upon LCD VDD power on (with in T2 max)-when the "Novideostream_Flag" (VB-ID Bit 3) is received from the source (at the end of T9).
-when no main link data, or invalid video data, is received from the source. Black video must be displayed within 64ms (typ) from the start of either condition. Video data can be deemed invalid based on MSA and timing information, for example.

Note 2: The sink may implement the ability to disable the black video function, as described in Note 1, above, for system development and debugging purpose.

Note 3: The sink must support AUX_CH polling by the source immediately following LCD VDD power on without causing damage to the sink device (the source can re-try if the sink is not ready). The sink must be able to respond to an AUX_CH transaction with the time specified within T3 max.

Display Port Panel B/L power sequence timing parameter:



	Min (ms)	Max (ms)
T13	0.5	10
T14	10	-
T15	0	-
T16	0	-
T17	10	-
T18	0.5	10
T19	1*	-
T20	1*	-

Seamless change: T19/T20 = 5xT_{PWM}*

*T_{PWM}= 1/PWM Frequency

Note 1 : If T14,T15,T16,T17<10ms，The display garbage may occur. We suggest T14,T15,T16,T17>10ms to avoid the display garbage.

Note 2：If T13 or T18<0.5ms，the inrush current may cause the damage of fuse. If T13 or T18<0.5ms，the inrush current I^2t is under typical melt of fuse Spec.，there is no mentioned problem.

7. Reliability Test Criteria

Environment test conditions are listed as following table.

Items	Required Condition	Note
Temperature Humidity Bias (THB)	Ta= 50℃, 80%RH, 300hours	
High Temperature Operation (HTO)	Ta= 60℃, 300hours	
Low Temperature Operation (LTO)	Ta= 0℃, 300hours	
High Temperature Storage (HTS)	Ta= 60℃, 300hours	
Low Temperature Storage (LTS)	Ta= -20℃, 300hours	
Vibration Test (Non-operation)	Acceleration: 1.5 Grms Wave: Random Frequency: 10 - 200 Hz Duration: 30 Minutes each Axis (X, Y, Z)	
Shock Test (Non-operation)	Acceleration: 50 G Wave: Half-sine Active Time: 20 ms Direction: ±X, ±Y, ±Z (one time for each Axis)	
Drop Test	Height: 46 cm, package test	
Thermal Shock Test (TST)	-20℃/30min, 60℃/30min, 100 cycles	*1*
On/Off Test	On/10sec, Off/10sec, 30,000 cycles	
ESD (Electro Static Discharge)	Contact Discharge: ± 8KV, 150pF(330Ω) 1sec, 8 points, 25 times/ point.	*2*
	Air Discharge: ± 15KV, 150pF(330Ω) 1sec 8 points, 25 times/ point.	

Note 1: The TFT-LCD module will not sustain damage after being subjected to 100 cycles of rapid temperature change. A cycle of rapid temperature change consists of varying the temperature from -20oC to 60oC, and back again. Power is not applied during the test. After temperature cycling, the unit is placed in normal room ambient for at least 4 hours before power on.

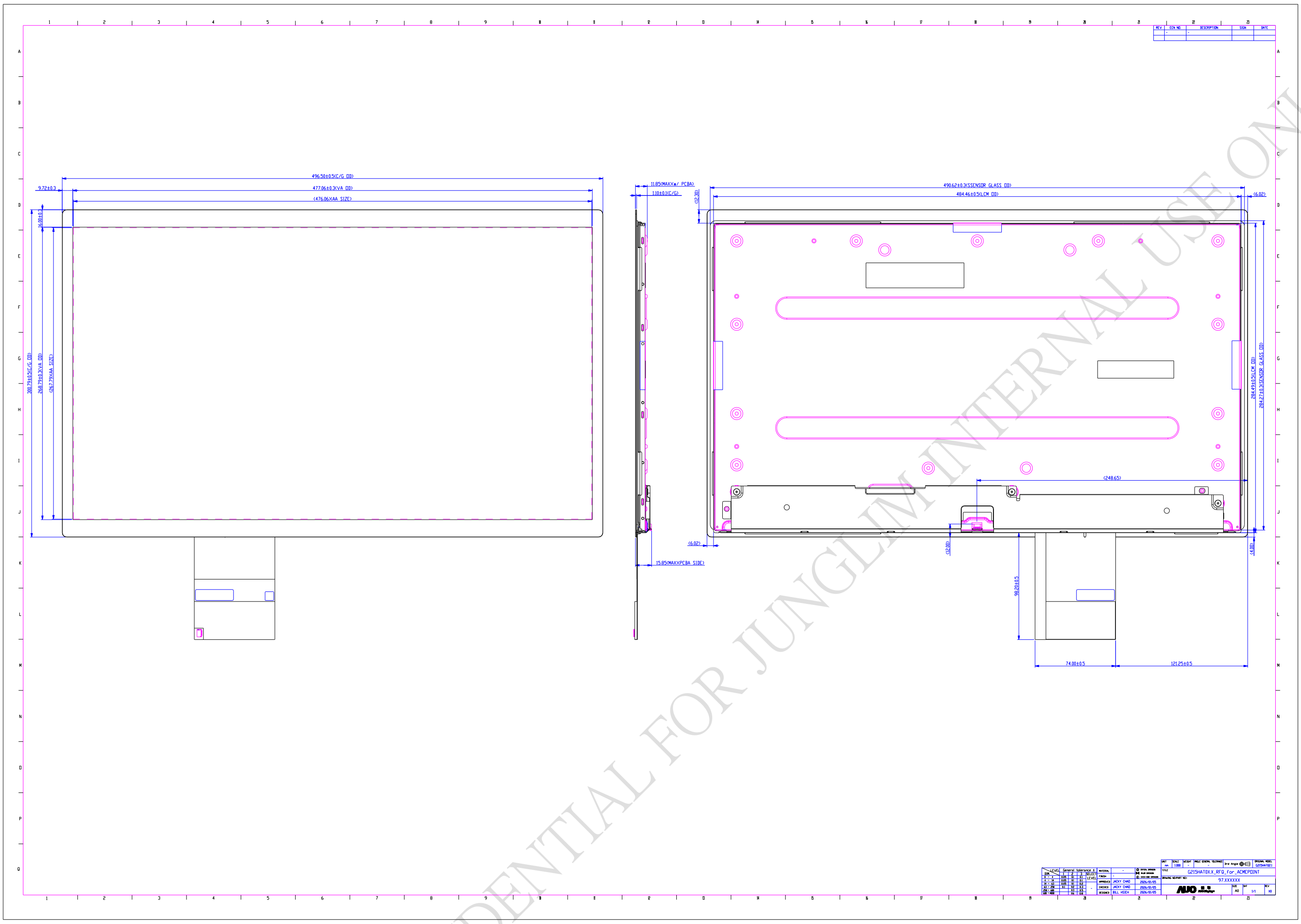
Note 2: According to EN61000-4-2 , ESD class B: Some performance degradation allowed. No data lost. Self-recoverable. No hardware failures.

Note 3: To inspect TFT-LCD module after reliability test, please store it at room temperature and room humidity for 24 hours at least in advance.

Note 4: The high temperature test item: THB & HTO & HTS & Thermal Shock, Mura shall be ignored.

8. Mechanical Characteristics

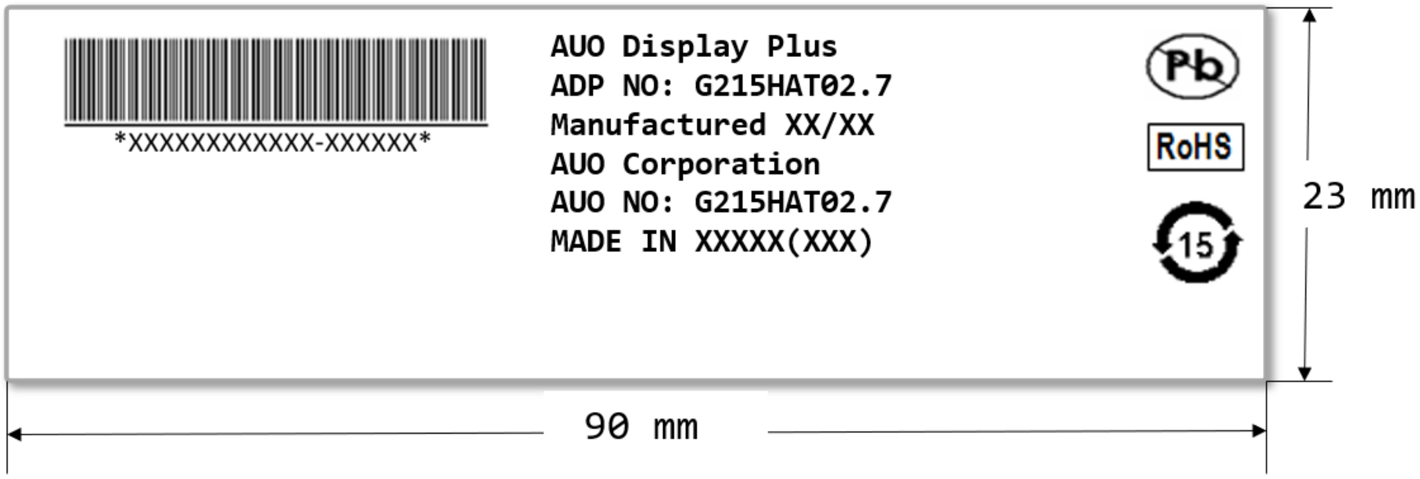
8.1 Total solution Outline Dimension



9. Label and Packaging

9.1 Shipping Label

The label is on the panel as shown below:



Note 1: For Pb Free products, AUO will add Pb for identification.

Note 2: For RoHS compatible products, AUO will add RoHS for identification.

Note 3: For China RoHS compatible products, AUO will add 15 for identification.
Note 4: The Green Mark will be presented only when the green documents have been ready by AUO Internal Green Team.

9.2 Carton Package
TBD

10. Safety

10.1 Keen Edge Requirements

There will be no sharp edges or comers on the display assembly that could cause injury.

10.2 Materials

10.2.1 Toxicity

There will be no carcinogenic materials used anywhere in the display module. If toxic materials are used, they will be reviewed and approved by the responsible AUO toxicologist.

10.2.2 Flammability

The printed circuit board will be made from material rated 94-V1 or better. The actual UL flammability rating will be printed on the printed circuit board.

10.3 Capacitors

If any polarized capacitors are used in the display assembly, provisions will be made to keep them from being inserted backwards.

10.4 International Safety Standard Compliance

The TFT-LCD module will satisfy all requirements for compliance to:IEC/UL 62368-1